

Project 4: Preference Elicitation

Human-Centric Artificial Intelligence

Design and Implementation Report

Movie preference elicitation using pairwise choices and ranked lists

September 2026

*This report describes the proposed study and the implemented interface.
The user study itself has not been conducted.*

1. Introduction

A recommender system is useful only when it can learn what a person likes. That becomes more difficult for a new user because there is no previous rating history to use. In this project, I focused on that “cold-start” situation and compared two simple ways of asking a new user about movie preferences.

The first method is a pairwise choice: the participant sees two movies and selects the one they would rather watch. The second method is a ranking task: the participant sees ten movies at the same time and orders them from most preferred to least preferred. Both interfaces are intended to estimate the same underlying preference model, but they ask the participant for information in very different ways.

The project uses the IMDB 5000 Movie Dataset. The dataset contains movie metadata rather than user ratings, so the study has to collect the preference information itself. I therefore needed to make two linked design decisions: first, how to represent each movie numerically; and second, how to turn a participant’s choices or rankings into an estimate of their preferences.

The report follows the four tasks in the project brief. Section 2 explains the movie features used by the recommender. Section 3 describes the Bradley-Terry model and its extension to complete rankings. Section 4 presents the proposed user study. Section 5 explains the participant interface and the usability decisions made for people who may have no technical background.

Scope of the project

The assignment asks for a rigorous study protocol and an interface that could be used to run the study. It does not ask for real participant recruitment or experimental results. For that reason, this report describes the planned analysis but does not invent any findings.

2. Task 1 - Representing a movie as features

The preference model uses a linear utility function, $U(x) = w^T x$. In practical terms, this means that every movie must be converted into a numerical vector x , while w represents how strongly a particular user cares about the different movie characteristics. The feature representation therefore needs to be informative enough to describe movie taste, but not so large that it becomes impossible to learn from a small number of interactions.

I chose a compact set of features that are easy to extract from the IMDB data and also easy to explain. This was important because the project is about a new user: with only a few choices, a model with thousands of actor, director, or title-specific variables would have too many parameters and very little evidence for most of them.

2.1 Features used

Feature	Encoding	Reason for including it
Genre	21 binary indicators	Genres are one of the clearest signals of movie taste. A user may consistently prefer genres such as Comedy, Horror, Drama, or Sci-Fi.
Release year	Standardized number	Allows the model to learn whether a user tends to prefer newer or older movies.

Duration	Standardized number	Captures whether the user tends to prefer shorter or longer films.
IMDb score	Standardized number	Provides a broad quality/popularity signal that is available in the dataset.
Language	English / non-English indicator	Adds a simple language preference signal without creating many very rare language categories.
Content rating	One-hot indicators	Allows preferences to differ across G, PG, PG-13, R, NC-17, and Unrated films.

I deliberately did not create separate features for every actor and director. Although those details can matter to real movie choices, the IMDB dataset contains many different names. One-hot encoding all of them would make the model very high-dimensional. With only a small number of preference tasks, many of those weights would never receive useful evidence. For this project, the smaller representation is a better match for the amount of feedback being collected.

2.2 Feature extraction

The extraction method is implemented in `services/feature_service.py`. The main steps are straightforward:

- Read the cleaned movie metadata and keep one row per movie.
- Split the genre field and create a 0/1 value for each of the 21 selected genres.
- Convert release year, duration, and IMDb score to numeric values.
- Replace a missing numeric value with the median of that column.
- Standardize the numeric fields so they are on comparable scales.
- Create the English-language indicator and the content-rating indicators.
- Concatenate all values into one feature vector for each movie.

For a numeric field, standardization is calculated as $(\text{value} - \text{mean}) / \text{standard deviation}$. This does not change which movies are high or low on that feature; it only changes the scale. Without this step, release year would contain numbers around 2000 while IMDb score is usually between 0 and 10, so the raw numerical sizes would be misleading during optimization.

Median imputation is used only to avoid dropping a movie because one numeric field is missing. It is a simple choice rather than an attempt to guess the true value. If this project were expanded into a real recommender, I would also test richer features such as plot embeddings, cast information, or collaborative signals, but the current representation is intentionally small and interpretable.

3. Task 2 - From two-movie choices to complete rankings

3.1 Bradley-Terry model for pairwise choices

For movie i , the model assigns a utility $U_i = \mathbf{w}^T \mathbf{x}_i$. A higher utility means that, according to the current preference vector, the movie is more attractive to that user. The Bradley-Terry model converts the difference between two utilities into a probability.

$$P(i \text{ preferred to } j) = \exp(U_i) / [\exp(U_i) + \exp(U_j)]$$

Every time the participant chooses one movie over another, the model receives a winner-loser pair. The preference vector $\mathbf{w}$ is estimated by maximizing the likelihood of those observed choices. The

implementation includes L2 regularization. I used this because the study intentionally collects only a small amount of data from each person, so regularization helps keep the estimated weights from becoming unnecessarily extreme.

3.2 Extending the model to a ranking

The second interface does not produce one winner and one loser. It produces a complete order such as $i1 > i2 > \dots > in$. Treating that ranking as only one pairwise comparison would throw away most of the information the participant provided. I therefore used the Plackett-Luce formulation, which can be understood as a sequence of choices from the remaining items.

For a ranking of ten movies, the first ranked movie is treated as the choice from all ten. After that movie is removed, the second ranked movie is treated as the choice from the remaining nine. The same idea continues until only one movie is left. The probability of the full ranking is the product of those sequential choice probabilities.

$$P(i1 > \dots > in) = \text{product over } k \text{ of } \exp(U_{ik}) / \text{sum of } \exp(U_{kj}) \text{ over the remaining movies}$$

This extension fits the project well for two reasons. First, it uses exactly the same movie utilities as the pairwise model, so the two elicitation interfaces remain comparable. Second, when only two items remain, the formula becomes the ordinary Bradley-Terry probability. In that sense, Plackett-Luce is a natural multi-item generalization of the pairwise model.

The ranking model is fitted by minimizing the negative log-likelihood with the same type of L2 penalty. The implementation also uses a log-sum-exp calculation in the denominator. That is mainly a numerical stability detail: it prevents overflow when utility values become large during optimization, without changing the mathematical model.

4. Task 3 - Proposed user study

4.1 Research question and hypotheses

The study asks a practical question: which interface lets a new user communicate movie preferences more effectively, while still being comfortable to use?

- Primary hypothesis (H1): the model learned from ranked lists will have higher agreement with the participant on held-out movie choices, because a complete ranking contains several relative preference relationships at once.
- Secondary hypothesis (H2): pairwise choices will receive a higher ease-of-use rating, because choosing between two movies is simpler than organizing ten movies.
- Exploratory measures: confidence ratings and time taken for each elicitation method. I would compare these outcomes but would not make a strong directional claim in advance.

4.2 Study design

I would use a within-subject design, meaning that every participant completes both interfaces. This is useful here because movie taste varies a lot from person to person. By testing both methods with the same participant, the comparison is less affected by differences in general movie knowledge or personal decisiveness.

To reduce order effects, the study has two sequences: AB (pairwise first, ranking second) and BA (ranking first, pairwise second). The demonstration chooses between them randomly for each session. In a

real study I would use controlled counterbalancing so the final sample contains the same number of participants in each order.

The two training conditions are matched on the number of unique movies shown. The pairwise condition contains 10 tasks with 2 movies each, giving 20 training movies. The ranking condition contains 2 tasks with 10 movies each, also giving 20 training movies. After both methods, participants complete 8 held-out pairwise choices using 16 additional movies. Sampling is without replacement within a session, so movies used for training are not reused for validation.

Why use separate validation movies?

The final validation choices are meant to test what the two learned models can predict about new decisions. If the same movies were used for both learning and validation, the accuracy would give an overly optimistic picture of model performance.

4.3 Participants and recruitment

For a real run of the study, I would recruit adults aged 18 or over who watch movies at least occasionally and can understand the instructions. Recruitment could use a university or community participant pool, mailing lists, and general volunteer channels. I would avoid recruiting only computer-science students, because the interface should work for non-technical users as well.

For the course protocol, a practical target would be at least 40 completed participants, with equal numbers in the AB and BA orders. This is a working target rather than a claim that 40 is statistically optimal. Before a confirmatory study, I would carry out an a-priori power analysis based on the chosen primary outcome and an expected effect size.

The study does not need names, student numbers, or similar personal identifiers. If persistent storage were added for real recruitment, each participant would receive a random session code instead.

4.4 Procedure

Stage	What the participant does
1. Information and consent	Read a short explanation of the study, data handling, voluntary participation, and the right to stop. Consent is collected before the study begins.
2. First elicitation method	Complete either the pairwise tasks or the ranking tasks, depending on the assigned order. Timing starts when the method begins and stops after its last task.
3. Short questionnaire	Rate ease of use and confidence on 1-5 scales, with an optional comment box.
4. Second elicitation method	Complete the other interface with a different set of movies. Its completion time is recorded separately.
5. Second questionnaire	Answer the same ease and confidence questions for the second method.
6. Held-out validation	Make 8 new pairwise choices. These answers are used only for evaluation, not for fitting either model.

7. Completion page	See the two agreement scores and completion times, together with a warning that one individual session cannot prove which method is scientifically better.
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4.5 Measures and planned analysis

The primary objective measure is predictive agreement on the same held-out pairwise decisions for both models. For each participant, I would calculate the proportion of validation choices that agree with the prediction from the pairwise-trained model and the ranking-trained model.

The subjective measures are ease of use and confidence, each collected on a 1-5 scale after the corresponding method. Completion time is recorded as a behavioral measure, and the optional text comment provides qualitative context when a participant found something confusing or particularly easy.

Because both conditions are completed by the same person, the main comparisons are paired. I would first inspect the distribution of participant-level differences. If a paired t-test is reasonably appropriate, I would use it; otherwise, I would use a Wilcoxon signed-rank test. I would follow the same paired principle for ease, confidence, and completion time. I would report the size of the difference and a confidence interval rather than relying only on a p-value.

I would also check whether the AB and BA orders behave differently. An order effect would not necessarily invalidate the experiment, but it would be important to report because the first interface could influence how a participant approaches the second one.

4.6 Ethics, privacy, and practical preparation

This is a low-risk study about movie preferences, but normal research principles still apply. Participation should be voluntary, participants should know they can stop, and the system should avoid collecting data that are not needed for the research question. For a real deployment, responses should be stored under deidentified session codes, access should be restricted, and a retention period should be defined.

Before recruiting participants, I would pilot the full study with several people who did not build the project. That pilot is especially important for the ranking interaction, because a control that feels obvious to the developer may not be obvious to somebody seeing it for the first time. I would also verify the mobile and keyboard experience, add persistent deidentified storage/export, enforce balanced AB/BA allocation, and freeze the analysis plan before looking at study outcomes.

5. Task 4 - Participant interface

The interface is implemented as a Django application. My main goal for the participant-facing pages was to avoid assuming any knowledge of recommender systems, probability models, or machine learning. A participant only needs to understand the immediate decision in front of them.

5.1 Landing page and navigation

The Project 4 landing page provides the two actions required by the brief: the user can download the project report or start the study. It also includes a short plain-language description of what the project is comparing. For an evaluator, expandable sections explain how Tasks 1-4 are covered without putting all of the technical detail in front of a participant at once.

Every study page has a visible way to return to the Project 4 home page. The project also includes a Main Home navigation button that returns to the main website. This matters because users should not feel trapped inside a multi-step study flow.

5.2 Help text and field explanations

I added expandable information panels for terms that may be unfamiliar. Instead of permanently filling the screen with definitions, a user can open a short explanation when needed. The same idea is used for movie fields such as year, genre, duration, content rating, IMDb score, and language.

This was a deliberate human-centered design choice. The project still exposes enough information to understand why two movies are different, but it does not require the participant to understand the internal feature vector. Technical terms such as “preference elicitation,” “utility,” and “held-out validation” are translated into ordinary language on the relevant pages.

5.3 Pairwise choice interface

In the pairwise condition, the participant sees two movie cards and answers one question: which movie would you rather watch? The page shows progress and uses the same metadata fields that appear in the ranking condition. When the participant makes a choice, the selected movie is stored as the winner and the other movie as the loser for the Bradley-Terry model.

5.4 Ranking interface

In the ranking condition, ten movies are shown in a numbered order. The participant moves them until the first position represents the movie they most want to watch and the last position represents the least preferred movie. Drag-and-drop is available, but I also added explicit Up and Down controls. This is important because not everybody knows that an item can be dragged, and drag-and-drop can be awkward on some devices or with keyboard navigation.

The application checks that all ten movies appear exactly once before accepting the ranking. This prevents an incomplete or duplicated ranking from being passed into the model.

5.5 Questionnaires, timing, and validation

After each elicitation method, the participant answers the same two rating questions: ease of use and confidence that the answers reflected their real preferences. An optional comment can be added. The elapsed time for the method is measured from the beginning of the method to completion of its final elicitation task, so time spent reading the initial consent page is not included.

The final validation section contains eight new two-movie choices. These responses are kept separate from model fitting. On the completion page, both learned preference vectors are evaluated against exactly the same validation decisions, which makes the displayed agreement scores directly comparable for that participant.

5.6 What the completion page means

The completion page reports how often each learned model agrees with the participant on the held-out choices, together with the time taken for the two elicitation methods. It can also show movie recommendations generated from the learned preferences. The page explicitly explains that these values describe one session only. A single participant cannot establish that one method is generally better; that conclusion would require the full study and the planned statistical analysis.

6. Limitations

The IMDB 5000 Movie Dataset is useful for this course project, but it is not a current or complete catalogue of movies. Its metadata may also reflect particular production and language distributions, so results from a real study would not automatically generalize to every audience.

The feature model is intentionally simple. It does not use plot text, posters, actor preferences, director preferences, or previous behavior from other users. This makes the model easier to explain and estimate from limited feedback, but it also means that some real reasons for liking a movie are not represented.

The current project is a study demonstration. Responses are kept in the Django session rather than in a research database, and the AB/BA order is randomly chosen independently for each session rather than balanced across a recruited sample. These are acceptable for demonstrating the required interface, but both would need to change before a real data-collection study.

7. Conclusion

This project compares two ways of learning a new user's movie preferences: repeated two-movie choices and complete rankings of ten movies. I represented each movie with a compact set of interpretable metadata features and used the Bradley-Terry model for pairwise choices. For ranked lists, I extended the same utility idea with a Plackett-Luce formulation so the full ordering contributes to the learned preference vector.

The proposed experiment uses a within-subject, counterbalanced design and evaluates both models on the same held-out movie choices. Ease, confidence, and completion time are collected as additional measures. The interface implements the complete study flow, including consent, both elicitation methods, questionnaires, validation, and a completion page.

The main interface changes were made with a non-technical participant in mind: short explanations are available when needed, movie fields are defined in ordinary language, ranking has button controls as well as dragging, and navigation back to the project home and main home is always available. The result is an interface that still implements the required preference models, but does not require the participant to understand those models in order to use it.